

Solution Requirements

Date	04 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Allow users/farmers to securely access the OptiCrop system (if login is implemented). Otherwise, users can directly access the crop prediction page.	High
2	Authorization Levels	Admin can manage datasets and ML models; farmers can submit soil parameters and view crop recommendations.	High
3	External Interfaces	Integrate the Flask backend with the trained Machine Learning model and agricultural dataset.	High
4	Transaction Processing	Accept Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall values; process them through the ML model to generate crop recommendations.	High

5	Reporting	Display the predicted crop recommendation and related information to the user.	Medium
S.No	Requirement Category	Requirement Description	Priority
6	Business Rules	Validate all input values before prediction and ensure the recommendation is generated only when all required parameters are provided.	High
7	Compliance	Protect user information and follow general data privacy practices.	Medium
8	Other	Provide a simple, responsive, and user-friendly interface for farmers.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	Category	Requirement Description	Target Metric
1	Performance & Speed	Generate crop recommendations quickly.	Response within 2–3 seconds
2	Scalability	Support multiple prediction requests without significant performance degradation.	Support at least 100 concurrent users
3	Security & Data Privacy	Validate inputs and protect user information from unauthorized access.	Secure input validation
4	Reliability & Availability	Ensure the application is available whenever users need crop recommendations.	99% uptime

5	Usability & Accessibility	Provide an easy-to-use interface accessible from desktop and mobile browsers.	User-friendly navigation
S.No	Category	Requirement Description	Target Metric
6	Maintainability	Modular Flask code that supports future updates and ML model improvements.	Easy to maintain and extend